

JavaScript Complete Map (From Scratch)

A zero-example, full-scope roadmap for mastering JavaScript.

Basics / Core

- JavaScript
- ECMAScript (ES5, ES6+)
- Interpreted language
- Single-threaded
- Synchronous vs Asynchronous
- Dynamic typing
- Case sensitivity
- Statements & expressions
- Comments

Variables

- var
- let
- const
- Naming rules
- Scope (global, block, function)
- Hoisting
- Temporal Dead Zone
- Shadowing
- Reassignment vs mutation

Data Types

- Primitive: String, Number, Boolean, Undefined, Null, BigInt, Symbol
- Non-Primitive: Object, Array, Function, Date, RegExp, Map, Set, WeakMap, WeakSet

Operators

- Arithmetic
- Assignment
- Comparison
- Logical
- Bitwise
- String operators
- Spread & Rest

- Optional chaining

Control Flow

- if
- else
- else if
- switch
- break
- continue
- return

Loops / Iteration

- for
- while
- do...while
- for...of
- for...in
- Iterators
- Generators
- Loop control keywords

Strings

- length
- charAt
- charCodeAt
- at
- concat
- includes
- indexOf
- lastIndexOf
- startsWith
- endsWith
- slice
- substring
- replace
- replaceAll
- split
- toUpperCase

- toLowerCase
- trim
- trimStart
- trimEnd
- repeat
- padStart
- padEnd
- match
- matchAll
- search
- localeCompare
- Template literals
- Escape characters

Arrays

- length
- push
- pop
- shift
- unshift
- splice
- slice
- concat
- join
- fill
- copyWithin
- reverse
- sort
- flat
- flatMap
- forEach
- map
- filter
- reduce
- reduceRight
- some
- every
- find

- findIndex
- findLast
- findLastIndex
- includes
- indexOf
- lastIndexOf
- Array.isArray

Objects

- Object literals
- Properties & methods
- Dot vs bracket notation
- Computed properties
- Getters & setters
- Object mutability
- Reference vs value
- Object.keys
- Object.values
- Object.entries
- Object.assign
- Object.freeze
- Object.seal
- Object.create
- Object.hasOwn

Functions

- Function declaration
- Function expression
- Arrow functions
- Parameters & arguments
- Default parameters
- Rest parameters
- Callbacks
- Higher-order functions
- Pure functions
- IIFE
- Recursion
- Closures

- this keyword
- Call, apply, bind

Async JavaScript

- Call stack
- Event loop
- Callback queue
- Microtask queue
- Callbacks
- Promises
- Promise states
- Promise chaining
- async
- await
- Async error handling

DOM & Web APIs

- DOM tree
- Nodes & elements
- Selecting & traversing
- Creating & modifying elements
- Attributes
- Classes
- Styles
- Events
- Bubbling
- Capturing
- Delegation
- Fetch API
- Storage APIs

Modules & JSON

- ES Modules
- import
- export
- Default export
- Named export
- Dynamic import

- JSON syntax
- JSON.stringify
- JSON.parse

Runtime & Ecosystem

- Browser JS
- Node.js
- Deno
- Bun
- NPM
- package.json
- Dependencies
- Bundlers
- Transpilers
- Linters
- Formatters